

GEOG 172 - Intermediate Geographical Analysis

Lecture Time: TR 12:30-1:45

Tutorial Time: T 5:00 - 6:50 & F 11:00 - 12: 50

Lecture Location: Ellison Hall 3621

Instructor: Peter Kedron

Office: Ellison Hall 5818

Office Hours: By appointment

Course Description:

GEOG 172 is designed to help students learn quantitative methods for the analysis of geographical data. Topics include spatial clustering, spatial auto-correlation, spatial regression, and introductory methods for analyzing point, area (lattice), and continuous data. Tutorial includes the use of statistical software for exploratory spatial data analysis.

Readings:

Course readings are provided through this canvas site. Lessons will be primarily drawn from two texts, which are also available electronically through the UCSB Library:

- Rogerson, P. (2021) *Spatial Statistical Methods for Geography*. Sage
- O'Sullivan, D., Unwin, D. (2010) *Geographic Information Analysis*. John Wiley and Sons, Hoboken: New Jersey.

Lecture materials will also draw selectively from the additional readings posted on the course and canvas website. Some additional readings will be required, but many are offered to help students delve deeper into topics they find particularly interesting.

Student Evaluation:

Evaluation will be based on tutorial assignments (85%) and participation (15%)

Tutorial Assignments: Students will complete seven tutorial assignments each worth 12 percent of their overall grade, which will account for 85 percent of their overall grade. Tutorial assignments involve a combination of work conducted in the weekly two hour tutorial session and independent work completing the assigned analysis and writing up the results. Students are encouraged to work with others when preparing their assignments. However, students must (1) submit their own assignments, (2) report who they help with the assignment, and (3) report who helped them with their assignment.

Tutorial assignments will be due at the start of the next tutorial session after which they were assigned (e.g., start of tutorial the following Thursday or Friday). Assignments will be submitted by uploading their materials (computational notebook, report, and data) to their corresponding Canvas assignment page. Late assignments will receive a 25% deduction for the overall potential grade for each day they are late (e.g., if you would have earned an 86%: 1 day -> a grade of 61%, 2 days -> a grade of 36%, ...).

For those unfamiliar with coding and computational notebooks, details about preparing and submitting assignments and template materials for those assignments will be presented in week 1.

Tutorial Schedule

Week	Due	Topic
0	—	No Tutorial
1	—	Prepare to Succeed in GEOG 172 (No assignment due for this tutorial)
2	3	Separate Northern and Southern California
3	4	Establish How Well Sampling Schemes Perform Across DGPs
4	5	Point Pattern Analysis
5	6	Examine the Openshaw Effect
6	7	Global Statistical Analysis
7	8	Local Statistical Analysis
8	10	Spatial Regressions
9	—	No Tutorial
10	—	No Tutorial

Class Participation: Participation in lecture and tutorial sessions will account for 15 percent of the student grades. Students are expected to come to class prepared and participate in class discussions and activities. Because this course asks you to develop and refine your conceptual understanding of quantitative analysis and your geographic analysis skills through activities, participation is essential to the learning process.

Students will be asked to participate in activities throughout the course. Activities are intended to serve as practice and checks on student knowledge. You are welcome and encouraged to work with your classmates on these exercises. Most will be done entirely in class. However, you are responsible for submitting your own individual solution report for each activity when requested.

Participation will be assessed in lecture and in tutorial. Participation will be evaluated throughout the course using the following scale:

- **15% — Fully Engaged Contributor.** Student comes to class prepared and contributes regularly without dominating. Contributions advance the conversation in substantive ways. For example, by connecting ideas across readings, raising productive complications, offering concrete examples from their own research domain, or helping the group work through a difficult concept. Shows genuine interest in and respect for others' perspectives. Actively participates in all group activities, including taking on different roles (e.g., facilitating, questioning, synthesizing) rather than defaulting to the same mode each time.

- ~10% — **Consistent Participant.** Student comes to class prepared and makes thoughtful contributions that reflect engagement with the material. Shows interest in and respect for others' views. Participates actively in small group work. Contributions are sound but tend to stay within the frame the readings or instructor have already established, rather than extending or challenging it.
- ~5% — **Present but Passive.** Student comes to class prepared and follows the discussion but contributes only minimally — for example, only when called upon or only in small group settings. Does not disrupt but does not help move the intellectual work of the class forward. May be absorbing ideas but is not yet making them visible to others.
- ~2% — **Underprepared.** Students come to class only partially prepared and contributes rarely. When contributions are made, they suggest limited engagement with the readings or activities. Participation in group work is inconsistent.

Grading Scheme: Grades in this course will be assigned using the scheme presented below.

A+	95%	B+	86.7-89.9%	C+	76.7-79.9%	D+	66.7-69.9%
A	92.5-94.9%	B	83.4-86.6%	C	73.4-76.6%	D	63.4-66.6%
A-	90.0-92.4%	B-	80.0-83.3%	C-	70.0-73.3%	D-	60.0-63.3%
						F	<60.0%

Topic Schedule

Readings will be provided as .pdf via the course website. R21 - Rogerson (2021) *Spatial Statistical Methods for Geography*. O10 - O'Sullivan and Unwin (2010) *Geographic Information Analysis*.

Week	Day	Date	Topic	Reading
0	R	9/24	Welcome to GEOG 172	—
1	T	9/29	Spatial Data the Foundation of Geographic Analysis	—
	R	10/1	<u>No Class</u> - Independent Statistics Review	R21-C2, O10-C1
2	T	10/6	Centrography	R21-C3
	R	10/8	Boundaries of Spatial Inequality	...
3	T	10/13	The Absence of Independence	O10-C4
	R	10/15	Sampling Spatial Data	Supplemental
4	T	10/20	Describing Point Patterns	O10-C5, R21-C4.6
	R	10/22	Assessing Point Patterns	O10-C5, RY8-C2.10
5	T	10/27	Scale and the Openshaw Effect	O10-C2.2, 7.2
	R	10/29	Reproducing the Openshaw Effect	Openshaw (1979)
6	T	11/3	Spatial Relations and Test Types	R21-C5, O10-C7
	R	11/5	Some Popular Global Statistics	...
7	T	11/10	Think Geographically, Measure Locally	R21-C6, O10-8
	R	11/12	Some Popular Local Statistics	...
8	T	11/17	Regression	R21-C8, O10-10
	R	11/19	Getting to Know Spatial Regression	Supplemental
9	T	11/24	Regression Lessons from the COVID Pandemic	Vijayan et al. (2021)
	R	11/26	<u>No Class</u> - Thanksgiving Break	
10	T	12/1	The Wider World of Geographical Analysis	Supplemental
	R	12/3	Emerging Topics in Geographical Analysis	...

Course and University Policies

Use of Artificial Intelligence Language Models:

Artificial intelligence (AI) language models, such as ChatGPT, may be used for code construction and refinement with appropriate citation, but not for the writing of assignments or projects. If you are in doubt as to whether you are using AI language models appropriately in this course, I encourage you to discuss your situation with me or the TAs. You are responsible for fact checking the code and statements composed by AI language models and the integrity of the content they generate - they have many strengths and many flaws.

At present, there are no universal guidelines for citing ChatGPT or other generative AI. If you use generative AI in this course, you must include (1) a statement that you did so, (2) provide a clear description of the content that was created with prompts (you may also include prompts and responses in an appendix), and (3) a citation to the model with name of the model or tool and version. Please see the example statement below:

ChatGPT (OpenAI 2023) was used to develop the code and conduct the analysis presented in this assignment. Specifically, ChatGPT was used to create R code to begin the development of the linear regression model using RStan. The prompts used are included below. A complete record of the code and text generated by ChatGPT is available in Appendix A.

Prompt 1: Generate R code to create a linear regression model using the package RStan

Prompt 2: Generate R code to test for spatial autocorrelation in the residuals of a linear regression model

OpenAI. (2023). ChatGPT (Version 4.0) [Large language model]. OpenAI. (Last accessed November 11, 2023). <https://www.openai.com/chatgpt>

It is your responsibility to ensure that your work meets the standards of this course, the UCSB Student Code of Conduct <https://www.sa.ucsb.edu/regulations/student-conduct-code/student-conduct-code>), and the wider scientific community.

Copyright:

All course materials (class lectures and discussions, handouts, examinations, web materials) and the intellectual content of the course itself are protected by United States Federal Copyright Law, the California Civil Code. The UC Policy 102.23 expressly prohibits students (and all other persons) from recording lectures or discussions and from distributing or selling lectures notes and all other course materials without the prior written permission of the instructor. Students are permitted to make notes solely for their own private educational use. Exceptions to accommodate students with disabilities may be granted with appropriate documentation. To be clear, in this class students are forbidden from completing study guides and selling them to any person or organization. The text has been approved by UC General Counsel. See <http://policy.ucop.edu/doc/2710530/PACAOS-100>.

Critical Student Support Services

- (1) Responsible scholarship Honesty and integrity in all academic work is essential for a valuable educational experience. The Office of Judicial Affairs has policies, tips, and resources for proper citation use, recognizing actions considered to be cheating or other forms of academic theft, and students' responsibilities, available on their website at: <http://judicialaffairs.sa.ucsb.edu>. Students are responsible for educating themselves on the policies and to abide by them.
- (2) Furthermore, for general academic support, students are encouraged to visit Campus Learning Assistance Services (CLAS) early and often. CLAS offers instructional groups, drop-in tutoring,

writing and ESL services, skills workshops and one-on-one consultations. CLAS is located on the third floor of the Student Resource Building, or visit <http://clas.sa.ucsb.edu>.

- (3) Managing stress / Supporting Distressed Students Personal concerns such as stress, anxiety, relationships, depression, cultural differences, can interfere with the ability of students to succeed and thrive. For helpful resources, please contact UCSB Counseling & Psychological Services (CAPS) at 805-893-4411 or visit <http://counseling.sa.ucsb.edu/>. If you encounter a student in distress, please contact 805-893-3030 immediately and/or consult the Responding to Distressed Student Protocol at <http://www.sa.ucsb.edu/distressedstudentsguide> or phone 893-3030.
- (4) Disabled Students Program: accommodations for exams Students with disabilities may request academic accommodations for exams online through the UCSB Disabled Students Program at <http://dsp.sa.ucsb.edu/>. Please make your requests for exam accommodations through the online system as early in the quarter as possible to ensure proper arrangement.
- (5) Sexual harassment. UCSB does not tolerate sexual harassment/sexual violence, which is prohibited by University policy and state and federal law. The Title IX Compliance and Sexual Harassment Policy Compliance Office (TIX/SHPC) provides assistance in preventing and resolving and investigating complaints of sexual harassment/sexual violence and gender discrimination. UCSB's Title IX office handles all issues surround sexual harassment: <https://titleix.ucsb.edu/>
- (6) Mandatory reporting. It is an instructor's responsibility to help create a safe learning environment on our campus. Students should be informed that instructors have a mandatory reporting responsibility related to their role as a professor. The instructor is *required* to share information regarding sexual misconduct or information about a crime that may have occurred on UCSB's campus or in the community. A result of the mandated report will be that students will receive outreach and resources from the campus Title IX office. Students may speak to someone confidentially by contacting CARE, Campus Advocacy, Resources & Education at the 24/7 advocacy line at (805) 893-4613 or visit them in person at the Student Resource Building.